

Analysing Social Engineering and Phishing Techniques

Tool: Social-Engineer Toolkit (SET)

Theory

Social engineering is the manipulation of people into revealing confidential information or performing actions that compromise security. The Social-Engineer Toolkit (SET) is an open-source framework included in Kali Linux designed to simulate social engineering attacks in authorised lab environments. This experiment demonstrates how a phishing page works by cloning a login page within an isolated local network only, strictly for educational awareness and security testing purposes.

Steps

1. Open the terminal in Kali Linux. Verify SET is installed. Run: `which setoolkit` — If not installed, run: `sudo apt install set -y`
 2. Find the Kali Linux IP address. Run: `ip addr show eth0` — Note the inet address, for example 192.168.56.101. This is the attacker IP used in the lab.
 3. Launch SET with root privileges. Run: `sudo setoolkit` — Accept the terms of service by typing y when prompted.
 4. From the SET main menu select option 1: Social-Engineering Attacks — Then select option 2: Website Attack Vectors — Then select option 3: Credential Harvester Attack Method — Then select option 2: Site Cloner
 5. Enter the Kali IP address when prompted for the listener IP. Enter the URL to clone when prompted, for example: `http://testphp.vulnweb.com` — SET clones the page and starts a web server on port 80.
 6. On another machine in the same lab network open a browser and navigate to the Kali IP address, for example: `http://192.168.56.101` — The cloned login page is displayed.
 7. Enter any test username and password on the cloned page and click login. Switch back to the Kali terminal. SET captures and displays the submitted credentials in plain text under the heading POSSIBLE USERNAME FIELD FOUND.
 8. Press Ctrl+C in the terminal to stop the SET web server after observing the captured output.
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Conclusion

SET successfully demonstrated how a cloned phishing page can capture login credentials entered by an unsuspecting user in a controlled lab environment. This experiment highlights the importance of verifying URLs before entering credentials and the need for security awareness training to defend against social engineering attacks.